

Deep reinforcement learning-based joint load scheduling for household multi-energy system

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ABSTRACT

Under the background of the popularization of renewable energy sources and gas-fired domestic devices in households, this paper proposes a joint load scheduling strategy for household multi-energy system (HMES) aiming at minimizing residents' energy cost while maintaining the thermal comfort. Specifically, the studied HMES contains photovoltaic, gas-electric hybrid heating system, gas-electric kitchen stove and various types of conventional loads. Yet, it is challenging to develop an efficient energy scheduling strategy due to the uncertainties in energy price, photovoltaic generation, outdoor temperature, and residents' hot water demand. To tackle this problem, we formulate the HMES scheduling problem as a Markov decision process with both continuous and discrete actions and propose a deep reinforcement learning-based HMES scheduling approach. A mixed distribution is used to approximate the scheduling strategies of different types of household devices, and proximal policy optimization is used to optimize the scheduling strategies without requiring any prediction information or distribution knowledge of system uncertainties. The proposed approach can handle continuous actions of power-shiftable devices and discrete actions of time-shiftable devices simultaneously, as well as the optimal management of electrical devices and gas-fired devices, so as to jointly optimize the operation of all household loads. The proposed approach is compared with a deep Q network (DQN)-based approach and a model predictive control (MPC)-based approach. Comparison results show that the average energy cost of the proposed approach is reduced by 12.17% compared to the DQN-based approach and 4.59% compared to the MPC-based approach.

1. Introduction

Nowadays more and more households use electricity, natural gas and renewable energy (such as small photovoltaic and wind power) as energy sources to meet multiple load demands, so as to reduce the household energy payment cost. This coupling system is called household multi-energy system (HMES) [1,2]. The system is bidirectionally grid-connected and the excess renewable generation can be sold to the grid. Since space heating and water heating are two major components of household energy consumption [3], many hybrid heating

technologies have been studied to develop more economical and flexible hybrid heating solutions for space and water heating, such as the combination of heat pump and gas boiler [4], the combination of gas-fired water heater and heat pump [5], and the combination of gas boiler, air conditioner and electric resistance water heater [6]. The hybrid heating system in HMES can bring economic benefits through switching energy forms (e.g., electricity and natural gas) or using more than one energy sources for heating. In addition, HMES also includes gas-electric kitchen stove (gas stove and induction cooker), washing machine, refrigerator and other household devices. In this context, how to

Abbreviations: AC, air conditioner; CREST, Center for Renewable Energy Systems Technology; DDPG, deep deterministic policy gradient; DQN, deep Q network; DRL, deep reinforcement learning; EB, electric boiler; GAE, generalized advantage estimation; GS, gas stove; HMES, household multi-energy system; HVAC, Heating, Ventilation, and Air Conditioning; IC, induction cooker; IES, integrated energy system; MDP, Markov decision process; MPC, model predictive control; PPO, proximal policy optimization; PV, photovoltaic; RH, resistance water heater; RL, reinforcement learning; TRPO, trust domain policy optimization; WGB, wall-hung gas boiler; WM, washing machine.

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schedule the various household loads to reduce energy cost while guaranteeing reliable customer comfort level has become an important topic in HMES research.

The multiple uncertainties existing in the system pose great challenges to the load scheduling of HMES. Specifically, these uncertainties include: (1) Uncertainty in residents' demands, such as hot water demand. (2) Uncertainty in energy prices, such as dynamic real-time electricity price. (3) Uncertainty in outdoor weather conditions, such as outdoor temperature. (4) Uncertainty in renewable energy generation, such as photovoltaic (PV) output and wind power. For household load scheduling/energy management considering uncertainty, it has received considerable attention in recent years. In [7], stochastic optimization and robust optimization were respectively used to determine the optimal operation of residential electrical appliances considering the uncertainty of real-time electricity price. In [8], a hybrid interval-stochastic method was proposed for residential energy management, in which the uncertainties in electricity price and PV generation were modeled by interval-based and stochastic scenarios. In [9], an electrical smart home energy management system was designed using mixed-integer quadratic-programming model predictive control (MPC). It predicted the future disturbance (occupancy information) to increase the scheduling performance. In [10], the household energy scheduling was formulated as a multi-objective MPC problem, where the objectives included energy cost, thermal comfort, carbon emission and user convenience. In [11], a rolling horizon optimization-based electricity scheduling method was proposed to tackle the uncertainties of renewable generation and electricity consumption.

The existing research mainly focus on the optimal scheduling of electrical devices, while the research for the scheduling of multiple energy sources in HMES are still in the initial stage [2]. [12] studied the optimal scheduling of electrical and thermal appliances in HMES, which used Weibull distribution to model the wind speed and used bimodal distribution to describe the solar irradiation. [13] proposed an operation optimization model of the residential hybrid energy system based on price response. [14] applied Monte Carlo simulation to model the electrical load uncertainty in residential hybrid energy system. [2] investigated the load scheduling problem of HMES equipped with gas-electric equipment, and used an interval optimization method to deal with the uncertainties by modeling uncertain parameters as interval numbers. [6] explored the advantage of using MPC to cope with system uncertainties in HMES over disregarding uncertainties in terms of cost saving. Although these research contribute significantly to the HMES scheduling, they depend strongly on modeling or predicting the system uncertainties. As it is difficult to obtain their accurate prediction or distribution knowledge, the scheduling results are greatly influenced by the accuracy of forecasting or mathematical model.

To avoid modeling or predicting HMES uncertainties in advance as [2,6,12–14] do and obtain more reliable scheduling strategy, reinforcement learning (RL) provides an effective solution to solve the stochastic decision-making problem. Markov decision process (MDP) theory-based RL has good decision making ability in the absence of initial knowledge of the environment [15] and has been used to solve a variety of problems including home/building energy management [16–18]. The work in [19] used Q-learning to make control decisions for Heating, Ventilation, and Air Conditioning (HVAC) and window systems to minimize both energy consumption and thermal discomfort. [20] proposed a multi-agent RL-based energy management method of residential appliances to minimize the energy bill and discomfort degree. The above approaches are characterized by the need to record optimization knowledge using Q-value table. However, the HMES consists of multitudinous controllable devices and the system state is continuous, the use of table-based RL approaches will lead to curse of dimensionality when the state or action space is too large [21], which makes the problem difficult to solve.

Deep RL (DRL) uses deep neural networks to approximate the function of RL, enabling RL to be extended to some previously difficult

problems, such as decision problems in continuous spaces [22]. In [23], deep Q network (DQN) and deep policy gradient methods were respectively used to learn the scheduling strategy of electrical devices for building energy management. In [24], a double DQN-based energy management method was proposed to perform scheduling of home appliances. It needs to discretize the consumption adjustment of the power-shift load, which significantly reduced the feasible action options. In [25], a home energy management approach was developed to schedule HVAC and electrical energy storage systems with continuous actions based on deep deterministic policy gradient (DDPG). The reviewed papers are summarized in Table 1.

While the aforementioned works have contributed to the applications of DRL technologies in household load scheduling, there are still two limitations of these approaches. First, they only deal with scheduling problems of domestic electric appliances, while it is necessary to consider the scheduling of both electricity and gas consumption in HMES to provide more flexibility and economy for customer energy usage with the popularity of gas-fired household devices. Second, there are various types of devices in HMES, such as air conditioner and wall-hung gas boiler that require continuous control actions, while washing machine and gas stove only have discrete on/off actions, which cannot be handled effectively by the above approaches.

To overcome the above limitations, we propose a DRL-based load scheduling approach to enable the joint scheduling of different types of devices in HMES, which is capable of handling both continuous and discrete actions of household electrical and gas devices. Since space heating and water heating account for a large proportion of household energy consumption, we only take the heating season as an example to carry out the research in this study. Case studies show that the gas-electric HMES can save 13.11% on average daily energy cost than pure electric solution. The innovative contributions of this paper are summarized as follows:

- A study of the load scheduling problem of HMES considering multiple optimization objectives is presented in this paper. The scheduling problem is converted to a Markov decision process containing both continuous actions of power-shiftable devices and discrete actions of time-shiftable devices, which takes into account the uncertainties of energy price, PV generation, outdoor temperature, and residents' hot water demand. It should be noted that the optimal scheduling of such a multi-energy system containing gas-electric hybrid heating system and gas-electric kitchen stove has not been studied in the existing works.
- To cope with the uncertainties in HMES and obtain a joint scheduling strategy of different types of household devices, a novel DRL-based approach is proposed. Specifically, we use a mixed distribution to approximate the scheduling policies of different types of loads, and then use proximal policy optimization (PPO) to optimize the scheduling policies. The proposed approach can simultaneously handle continuous actions and discrete actions, and perform the optimal management of gas-electric devices to jointly optimize the operation of all different kinds of household devices, which provides a new solution to the HMES load scheduling problem and a good basis for expanding its application in larger scales.
- The proposed approach can obtain adaptive scheduling actions based on the current observed state, which requires no distribution knowledge or prediction information of system uncertainties. In this sense, the proposed approach can successfully avoid the unreliability of the scheduling strategy caused by inaccurate prediction or modeling, and has better implementability than traditional approaches which rely on prediction or distribution knowledge of system uncertainties.

The remainder of this paper is organized as follows. Section 2 provides a mathematical description of the load scheduling problem of HMES. Section 3 converts the above problem to an MDP and gives a

Table 1
Summary of the reviewed papers on household energy optimization.

Ref	Energy system	Energy resources	Objective	Uncertainty	Approach
[2]	HMES	Electricity, gas, solar	Energy cost, emissions cost	PV generation, hot water demand and temperature, ambient and room temperature	Interval optimization
[6]	HMES	Electricity, gas, solar	Energy cost, thermal comfort	PV generation, inflexible load, outdoor temperature, hot water demand	MPC
[7]	Residential appliances	Electricity	Electricity payment	Electricity price	Stochastic optimization, robust optimization
[8]	Smart home	Electricity, solar	Expected profit	PV generation, electricity price	Hybrid interval stochastic approach
[9]	Smart home	Electricity, solar	Thermal comfort, monetary cost, energy efficiency	Solar irradiation, ambient temperature, occupancy	MPC
[10]	Smart home	Electricity, solar	Energy cost, thermal comfort, carbon emission, user convenience	Uncontrollable load, hot water draw, weather	MPC
[11]	Smart home	Electricity, solar, wind	Electricity cost payment	Renewable generation, electricity consumption	Rolling horizon optimization
[12]	HMES	Electricity, gas, solar, wind	Energy cost, operation and maintenance cost	Renewable generation	Distribution knowledge-based
[14]	HMES	Electricity, gas	Operation cost	Electrical load	Monte Carlo simulation
[20]	Residential appliances	Electricity	Energy bill, discomfort degree	Electricity price, load demand	Neural network and multi-agent RL
[24]	Smart home	Electricity, solar	Electricity cost	Users' behavior	DQN, double DQN
[25]	Smart home	Electricity, solar	Energy cost, battery depreciation cost, thermal comfort	Renewable generation, electricity price, outdoor temperature, non-shiftable demand, thermal model	DDPG

detailed DRL solution scheme. Simulation results are given in Section 4 to verify the effectiveness of the proposed approach. Finally, Section 5 concludes the paper.

2. System modeling and problem formulation

A typical HMES integrates electricity, natural gas and onsite renewable energy generation as its energy sources. Taking the HMES shown in Fig. 1 as an example, it uses roof-top PV systems as renewable energy sources. Household devices can be divided into time-shiftable loads, power-shiftable loads and non-shiftable loads according to their operation characteristics [26,27]. Power-shiftable loads represent these who can adjust the working power continuously within the predefined range. Time-shiftable loads represent these who can shift the working

time within a preferred working period [26]. Non-shiftable loads represent these whose demands must be satisfied and cannot be scheduled. Specifically, the power-shiftable loads include air conditioner (AC), wall-hung gas boiler (WGB) and resistance water heater (RH); time-shiftable loads include washing machine (WM), gas stove (GS) and induction cooker (IC); non-shiftable loads include refrigerator, alarm system, lighting, television and other electronic devices which cannot be scheduled. In this study, the refrigerator and alarm system are studied as examples of non-shiftable loads. AC, WGB and RH constitute the gas-electric hybrid heating system; GS and IC constitute the gas-electric kitchen stove.

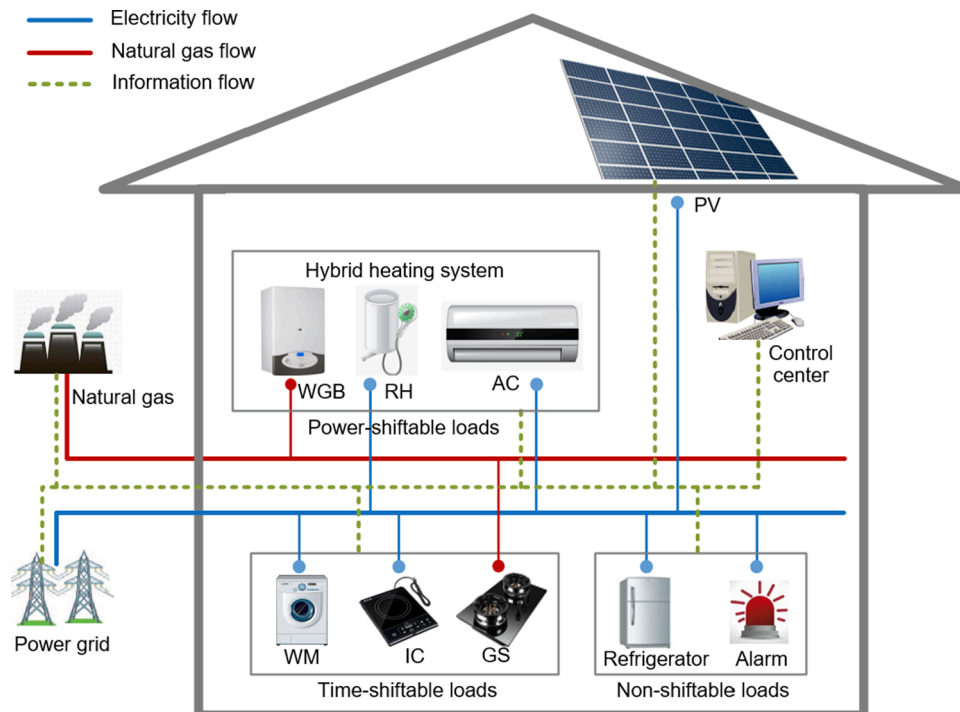


Fig. 1. Structure of the household multi-energy system.

2.1. Gas-electric hybrid heating system modeling

The gas-electric hybrid heating system is modeled in this subsection. We consider a hybrid heating system consisting of a WGB, a RH and an AC as described in [6], in which the WGB is physically connected to a water storage tank with a RH installed. The WGB consumes gas for both space heating and hot water production. It has two separate loops: a water heating loop and a space heating loop. The WGB takes cold water and outputs hot water to the water storage tank in the water heating loop. The heating medium which is output from the WGB flows through the radiators to heat the space in the space heating loop. The AC and RH consume electricity for space heating and water heating, respectively. Therefore, the space heating can be provided by WGB or/and AC; the water heating can be provided by WGB or/and RH. The WGB, RH and AC are power-shiftable loads, i.e., their working power is changeable and they can operate flexibly in predefined ranges.

(1) Space heating

For the AC, the generated heat energy Q'_{AC} can be expressed as.

$$Q'_{AC} = \eta_{AC} \cdot P'_{AC} \quad (1)$$

where P'_{AC} is the electric power consumption of AC at time step t , η_{AC} is the heating efficiency which may take different values in different seasons.

For the WGB, the heat exchange from the radiators to the indoor environment at time step t is expressed as [6].

$$Q'_{rad} = n \cdot Q'_{rad}^{unit} \cdot (T_{med}^t - T_{in}^t) / \Delta T_s \quad (2)$$

where n is the number of the radiators, Q'_{rad}^{unit} is the heat exchange power of a radiator under the standard temperature difference, T_{in}^t is the indoor temperature at time step t , T_{med}^t is the medium temperature in the radiator at time step t , and ΔT_s is the standard excess temperature. The temperature evolution of the medium in the radiator can be modeled as [6].

$$T_{med}^{t+1} = T_{med}^t + \frac{(Q'_{WGB_Space} - Q'_{rad}) \cdot \Delta t}{M_{med} \cdot C_{med}} \quad (3)$$

where Q'_{WGB_Space} is the heating power of WGB for space heating at time step t , M_{med} is the mass of the heating medium; C_{med} is the specific heat capacity of the medium; Δt is the time length of one time step.

Considering the space heating power generated by AC and WGB and the heat exchange between the indoor and outdoor air, the indoor temperature evolution can be modeled as.

$$T_{in}^{t+1} = \exp[-\Delta t / (RC)] T_{in}^t + \{1 - \exp[-\Delta t / (RC)]\} [R(Q'_{AC} + Q'_{rad}) + T_{out}^t] \quad (4)$$

where T_{out}^t is the outdoor temperature at time step t , R is the thermal resistance of the house and C is the thermal capacity of the indoor air. Note that the windows and doors openings and any type of air infiltration into the house are neglected.

The AC and WGB should satisfy the following operation constraints:

$$0 \leq P'_{AC} \leq P_{AC}^{max} \quad (5)$$

$$0 \leq Q'_{WGB_Space} \leq Q_{WGB_Space}^{max} \quad (6)$$

$$T_{med}^{min} \leq T_{med}^t \leq T_{med}^{max} \quad (7)$$

(2) Domestic hot water production

The water is heated by WGB or/and RH. The water will flow into the

water storage tank immediately after being heated by WGB. For WGB, the power for water heating is defined as:

$$Q'_{WGB_Water} = C_{water} M'_{WGB} (T_{WGB_set} - T_{cold}) / \Delta t \quad (8)$$

where Q'_{WGB_Water} is the water heating power of WGB at time step t , C_{water} is the specific heat capacity of water, T_{WGB_set} is the temperature set-point of the hot water, M'_{WGB} is the mass of the hot water heated by WGB and flowing to the water storage tank at time step t , T_{cold} is the temperature of the cold water, which is treated as a fixed value during the heating season.

The water temperature in the water storage tank is mainly affected by the following factors: the heat contribution due to the hot water flowing from the WGB, the heating power of the RH, the replenishing of cold water, and the heat loss due to hot water usage [6]. It is noted that the heat losses in the water storage tank by its non-perfect insulation are neglected. Therefore, the thermal environment in the water storage tank can be modeled as.

$$M_{tank} C_{water} T_{water}^{t+1} = M_{tank} C_{water} T_{water}^t + M'_{WGB} C_{water} T_{WGB_set} + Q'_{RH} \Delta t + M'_{cold} C_{water} T_{cold} - M'_{use} C_{water} T_{water}^t \quad (9)$$

where T_{water}^t is the water temperature in the storage tank at time step t ; M_{tank} is the mass of the water storage tank; Q'_{RH} is the heating power of RH at time step t ; M'_{use} is the mass of hot water usage at time step t ; M'_{cold} is the mass of cold water flowing into the water storage tank at time step t , which is equal to the hot water usage minus the hot water supplied by WGB, as described in Eq. (10):

$$M'_{cold} = M'_{use} - M'_{WGB} \quad (10)$$

According to Eqs. (9) and (10), the water temperature in the water storage tank can be modeled as:

$$T_{water}^{t+1} = T_{water}^t + \frac{M'_{WGB}}{M_{tank}} T_{WGB_set} + \frac{Q'_{RH} \Delta t}{M_{tank} C_{water}} + \frac{(M'_{use} - M'_{WGB})}{M_{tank}} T_{cold} - \frac{M'_{use}}{M_{tank}} T_{water}^t \quad (11)$$

The heating power of WGB, the electric power consumption of RH and the mass of the water flowing from WGB into the storage tank should be within certain ranges, which can be expressed as.

$$0 \leq Q'_{WGB_Water} \leq Q_{WGB_Water}^{max} \quad (12)$$

$$0 \leq P'_{RH} \leq P_{RH}^{max} \quad (13)$$

$$0 \leq M'_{WGB} \leq M_{WGB}^{max} \quad (14)$$

where P'_{RH} is the electric power of RH at time step t , and has the following relationship.

$$Q'_{RH} = P'_{RH} \cdot \eta_{RH} \quad (15)$$

where η_{RH} is the heating efficiency of RH.

The natural gas consumption of WGB at time step t can be expressed as follows:

$$V'_{WGB} = (Q'_{WGB_Space} + Q'_{WGB_Water}) / (\eta_{WGB} H_{gas}) \quad (16)$$

where η_{WGB} is the efficiency of WGB, H_{gas} is the energy contained in natural gas per volume.

2.2. Other devices modeling

(1) Washing machine

The WM is a time-shiftable device, i.e., it can shift the working time within a certain time period. The model of WM is expressed as.

$$\sum_{t=t_{WM,b}}^{t_{WM,e}} c_{WM}^t = L_{WM} \quad (17)$$

$$c_{WM}^t = \begin{cases} 0, & \forall t \in [1, t_{WM,b}) \cup (t_{WM,e}, T] \\ 0 \text{ or } 1, & \forall t \in [t_{WM,b}, t_{WM,e}] \end{cases} \quad (18)$$

$$\sum_{m=m}^{m+L_{WM}-1} c_{WM}^m \geq (c_{WM}^m - c_{WM}^{m-1}) \cdot L_{WM}, \forall m \in (t_{WM,b}, t_{WM,e} - L_{WM} + 1], m \in T \quad (19)$$

$$P_{WM}^t = c_{WM}^t \cdot P_{WM} \quad (20)$$

where c_{WM}^t represents the working status (on or off) of WM at time step t ; $t_{WM,b}$ and $t_{WM,e}$ represent the starting time and deadline of WM, respectively; L_{WM} represents the number of time steps that WM needs to run to fulfill a laundry task; T is the total time steps throughout the scheduling horizon; P_{WM} is the rated power of the WM; P_{WM}^t is the electric power consumption of the WM at time step t . Eq. (19) ensures that the WM will not be interrupted once started until the task is completed [28].

(2) Gas-electric kitchen stove

The cooking time is divided into morning, noon and evening. During each runtime range, the cooking process can be shifted within a certain time period. Once it starts working, the cooking process must be continuous. The mathematical model of KS is expressed as.

$$\sum_{t=t_{KS,b}}^{t_{KS,e}} c_{KS}^t = \frac{1}{3} L_{KS} \quad (21)$$

$$c_{KS}^t = \begin{cases} 0, & \forall t \notin [t_{KS,b}, t_{KS,e}] \\ 0 \text{ or } 1, & \forall t \in [t_{KS,b}, t_{KS,e}] \end{cases} \quad (22)$$

$$\sum_{m=m}^{m+\frac{1}{3}L_{KS}-1} c_{KS}^m \geq (c_{KS}^m - c_{KS}^{m-1}) \cdot \frac{1}{3} L_{KS}, \forall m \in \left(t_{KS,b}, t_{KS,e} - \frac{1}{3} L_{KS} + 1 \right], m \in T \quad (23)$$

$$V_{GS}^t = c_{GS}^t \cdot P_{KS} / (\eta_{GS} \cdot H_{gas}) \quad (24)$$

$$P_{IC}^t = c_{IC}^t \cdot P_{KS} \quad (25)$$

$$c_{GS}^t \cdot c_{IC}^t = 0 \quad (26)$$

where c_{KS}^t is the execution state of the cooking task at time step t ; c_{GS}^t and c_{IC}^t represent the working status of GS and IC, respectively; $c_{KS}^t = c_{GS}^t + c_{IC}^t$; $t_{KS,b}$ and $t_{KS,e}$ represent the starting time and deadline of the KS operation, respectively; L_{KS} denotes the number of total time steps that KS needs to run in the morning, noon and evening. P_{KS} is the rated power of the cooking; V_{GS}^t is the natural gas consumption of GS at time step t ; η_{GS} is the energy conversion efficiency of GS; P_{IC}^t is the electric power consumption of IC at time step t . Eq. (26) forces that IC and GS cannot be used at the same time.

(3) Non-shiftable devices

Since the schedules of non-shiftable devices such as refrigerator and alarm system cannot be changed and they are in operation within pre-defined ranges, their power consumption P_{must}^t can be expressed as known input data at time step t .

2.3. Optimization objective of the problem

The objective of the joint scheduling of HMES is to minimize the energy cost (including electricity cost and natural gas cost) and resi-

dents' thermal comfort loss through the optimal scheduling of household devices. The optimization is done sample by sample during the day in an operational way, with a 15-mins time step. The objective function is defined as follows:

$$F = \min \sum_{t=1}^T \left(w_1 C_1 \left(P_{grid}^t, V_{gas}^t \right) + w_2 C_2 \left(T_{in}^t \right) + w_3 C_3 \left(T_{water}^t \right) \right) \quad (27)$$

where w_1 , w_2 and w_3 are weighting factors set according to user preferences to achieve the trade-off between energy cost and thermal discomfort; $C_1 \left(P_{grid}^t, V_{gas}^t \right)$ is the sum of electricity cost of electrical devices and natural gas cost of gas-fired devices, $C_2 \left(T_{in}^t \right)$ is the penalty caused by indoor temperature discomfort and $C_3 \left(T_{water}^t \right)$ is the penalty caused by hot water temperature discomfort, which are defined as.

$$C_1 \left(P_{grid}^t, V_{gas}^t \right) = \left(\frac{\epsilon_{pu}^t + \epsilon_{sc}^t}{2} \cdot P_{grid}^t + \frac{\epsilon_{pu}^t - \epsilon_{sc}^t}{2} \cdot |P_{grid}^t| \right) \cdot \Delta t + \epsilon_{gas}^t \cdot V_{gas}^t \cdot \Delta t \quad (28)$$

$$C_2 \left(T_{in}^t \right) = \sigma_1 |T_{in}^t - T_{in}^{pre}| / (T_{in}^{max} - T_{in}^{min}) \quad (29)$$

$$C_3 \left(T_{water}^t \right) = \sigma_2 |T_{water}^t - T_{water}^{pre}| / (T_{water}^{max} - T_{water}^{min}) \quad (30)$$

where P_{grid}^t is the electric power exchange between the HMES and the power grid at time step t , which means the HMES purchases electric power from the grid when it is positive, and sells electric power to the grid when it is negative; V_{gas}^t is the total natural gas consumption; ϵ_{pu}^t is electricity purchasing price and ϵ_{sc}^t is the electricity selling price; ϵ_{gas}^t is natural gas price; the factors σ_1 and σ_2 are introduced to map the indoor temperature discomfort and water temperature discomfort levels into money in Eqs. (29) and (30), respectively, which are set based on the survey on customers who understand 'costs in monetary units' [6]; T_{in}^{pre} is the satisfactory indoor temperature setting value; T_{water}^{pre} is the hot water temperature setting value; T_{in}^{min} and T_{in}^{max} are the lower bound and upper bound of indoor temperature, respectively; T_{water}^{min} and T_{water}^{max} are the lower bound and upper bound of water temperature in the water tank, respectively. P_{grid}^t and V_{gas}^t are denoted as.

$$P_{grid}^t = P_{must}^t + P_{AC}^t + P_{RH}^t + P_{WM}^t + P_{IC}^t - P_{PV}^t \quad (31)$$

$$V_{gas}^t = V_{WGB}^t + V_{GS}^t \quad (32)$$

where P_{PV}^t is the PV generation at time step t . The constraints of the HMES joint scheduling problem are presented in Eqs. (1)-(26).

3. Proposed DRL-based joint load scheduling approach

3.1. Conversion of the problem to a Markov decision process

The load scheduling problem of the HMES is essentially a sequential decision problem, which can be formulated as a Markov decision process (MDP). In this subsection, the studied problem is converted to an MDP. The MDP includes several important elements: environment, agent, state, action and reward function [29]. The agent observes the environment state s_t at time step t , and performs action a_t . Then it receives the immediate reward r_t from the environment and the environment evolves to the next state s_{t+1} . The interaction process between the agent and the environment is shown in Fig. 2.

Environment: The environment refers to the household multi-energy system in this study, as shown in Fig. 1.

Agent: The agent refers to the energy management system of the HMES, also known as the control center in this paper. The trained agent can develop optimal load scheduling strategy according to the state of the environment.

State: The state represents the known environment observed by the

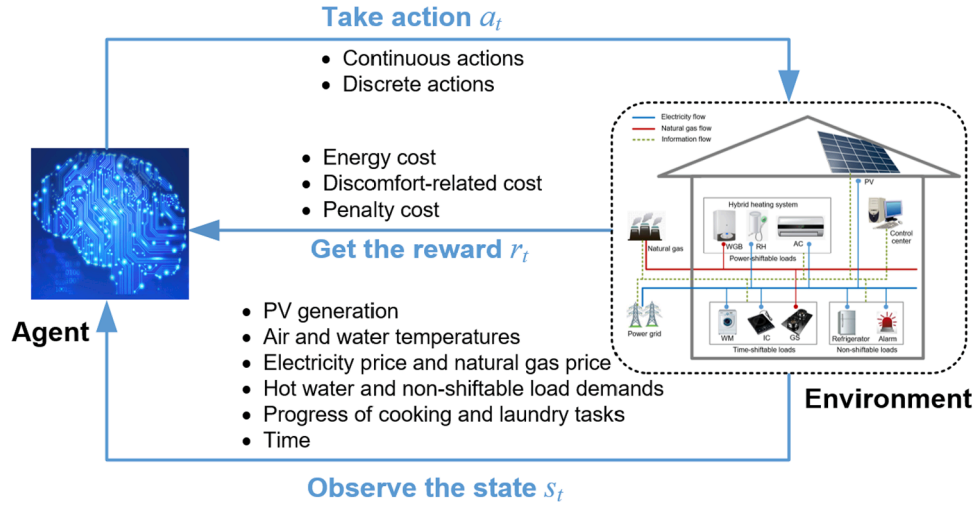


Fig. 2. The interaction between the agent and the environment in the MDP.

agent. In the process of HMES load scheduling, the environment state observed by the agent at time step t includes PV generation, power consumption of non-shiftable devices, indoor temperature, outdoor temperature, hot water temperature, hot water demand, electricity price (including electricity purchasing price and electricity selling price), natural gas price, the progress of laundry task, the progress of cooking task, and the physical time. Generally, the electricity selling price is related to the electricity purchasing price and can be calculated from the electricity purchasing price [30], so it is not included in the state. The environment state at time step t can be expressed as.

$$s_t = [P_{PV}^t, P_{must}^t, T_{in}^t, T_{out}^t, T_{water}^t, M_{use}^t, e_{pu}^t, e_{gas}^t, \rho_{WM}^t, \rho_{KS}^t, t] \quad (33)$$

where ρ_{WM}^t and ρ_{KS}^t can be described as.

$$\rho_{WM}^t = \begin{cases} \sum_{k=t_{WM,b}}^{t-1} c_{WM}^k / L_{WM}, & \forall t \in [t_{WM,b}, t_{WM,e}] \\ 0, & \text{otherwise} \end{cases} \quad (34)$$

$$\rho_{KS}^t = \begin{cases} \sum_{k=t_{KS,b}}^{t-1} c_{KS}^k / \frac{1}{3} L_{KS}, & \forall t \in [t_{KS,b}, t_{KS,e}] \\ 0, & \text{otherwise} \end{cases} \quad (35)$$

Action: The control center makes the load scheduling decision according to the observed environment state. In terms of the HMES load scheduling problem, the decision variables include the switching actions of WM, IC and GS (time-shiftable devices), the space heating power and water heating power of WGB, the electric power of RH, and the electric power of AC (power-shiftable devices). Since the non-shiftable devices cannot be scheduled, they are always “on” within predefined ranges. Therefore, the action of the agent can be defined as.

$$a_t = [c_{WM}^t, c_{IC}^t, c_{GS}^t, Q_{WGB_Space}^t, Q_{WGB_Water}^t, P_{RH}^t, P_{AC}^t] \quad (36)$$

Reward function: The agent receives reward r_t after performing the action a_t . The objective of the HMES load scheduling is to minimize the total cost defined by Eq. (27). To achieve this goal, the reward function at time step t is defined as.

$$r_t(s_t, a_t) = -[w_1 C_1(s_t, a_t) + w_2 C_2(s_t) + w_3 C_3(s_t)] - C_4(s_t, a_t) \quad (37)$$

$$C_4(s_t, a_t) = \begin{cases} 0, & \text{if } T_{in}^t \in [T_{in}^{\min}, T_{in}^{\max}] \text{ and } T_{water}^t \in [T_{water}^{\min}, T_{water}^{\max}] \text{ and } c_{GS}^t \cdot c_{IC}^t = 0 \\ \xi, & \text{otherwise} \end{cases} \quad (38)$$

where $C_1(s_t, a_t)$ is the energy cost at time step t in Eq. (28); $C_2(s_t)$ is the penalty caused by indoor temperature discomfort in Eq. (29); $C_3(s_t)$ is

the penalty caused by water temperature discomfort in Eq. (30); $C_4(s_t, a_t)$ is the penalty term when indoor temperature or hot water temperature is beyond the defined range or IC and GS work at the same time; ξ is a positive value. Since the agent generally takes the reward maximization form, the minus sign is used at the front of the right side of Eq. (37) to take the opposite of the total cost.

The objective of the MDP is to discover the optimal load scheduling policy π^* to maximize the expectation of the discounted cumulative rewards, which can be described as.

$$\max_{\pi \in \Pi} J(\pi) = \mathbb{E}_{\pi} \left[\sum_{t=1}^T \gamma^{t-1} r_t \right] \quad (39)$$

where $\mathbb{E}_{\pi}$ is the expected value under policy π , Π is the set of all admissible policies, and the parameter $\gamma \in [0, 1]$ is the discount factor.

3.2. Deep neural network-based policy

In Section 3.1, the action of the MDP is formulated as the decision variables of time-shiftable devices and power-shiftable devices, where the former are discrete quantities and the latter are continuous quantities, i.e.

$$a_t = \underbrace{[c_{WM}^t, c_{IC}^t, c_{GS}^t]}_{\text{time-shiftable}}, \underbrace{[Q_{WGB_Space}^t, Q_{WGB_Water}^t, P_{RH}^t, P_{AC}^t]}_{\text{power-shiftable}} \quad (40)$$

Considering that there are both discrete and continuous quantities in the action, a mixed probability distribution is used to jointly approximate the load scheduling policy π , denoted as.

$$\pi \sim \begin{cases} \mathcal{B}(p_{ts}(s_t)), & \text{if } a_t \in \{c_{WM}^t, c_{IC}^t, c_{GS}^t\} \\ \mathcal{N}(\mu_{ps}(s_t), \sigma_{ps}^2), & \text{otherwise} \end{cases} \quad (41)$$

where Bernoulli distribution $\mathcal{B}(p_{ts}(s_t))$ is used to approximate the policy for discrete actions of time-shiftable devices (WM, IC and GS), $p_{ts}(s_t)$ denotes the probability of enabling time-shiftable device ts to work; Gaussian distribution $\mathcal{N}(\mu_{ps}(s_t), \sigma_{ps}^2)$ is used to approximate the policy for continuous actions of power-shiftable devices (WGB, RH and AC), $\mu_{ps}(s_t)$ denotes the mean of power-shiftable device ps and σ_{ps} is the standard deviation.

In order to determine the parameters of Bernoulli distribution and Gaussian distribution, i.e., $p_{ts}(s_t)$, $\mu_{ps}(s_t)$ and σ_{ps} , a deep neural network is used to learn them. The inputs of the neural network are the environment state, i.e., PV generation, power consumption of non-shiftable

loads, indoor temperature, outdoor temperature, hot water temperature, hot water demand, electricity price, natural gas price, progresses of the laundry task and cooking task and the physical time; the outputs of the network are Bernoulli distribution parameters $p_{ts}(s_t)$ for discrete actions of time-shiftable devices, and Gaussian distribution parameters $\mu_{ps}(s_t)$ and logarithmic standard deviation $\log(\sigma_{ps})$ for continuous actions of power-shiftable devices. The neural network can be represented as follows.

$$p_{ts}(s_t; \theta) = \text{Sigmoid}([W]_{ts} \cdot f(s_t) + [b]_{ts}) \quad (42)$$

$$\mu_{ps}(s_t; \theta) = [W]_{ps+3} \cdot f(s_t) + [b]_{ps+3} \quad (43)$$

$$\log \sigma_{ps}(\theta) = [W]_{ps} \quad (44)$$

where $W, W_\sigma, b \in \theta$ represent the weight matrices and bias vectors of the output layer, respectively; $\text{Sigmoid}(\cdot)$ is the sigmoid activation function; $f(s_t)$ is expressed as.

$$f(s_t) = \text{LReLU}(W_L \cdot v_L(s_t) + b_L) \quad (45)$$

$$v_{l+1}(s_t) = \text{LReLU}(W_l \cdot v_l(s_t) + b_l), l = 1, \dots, L-1 \quad (46)$$

$$v_1(s_t) = s_t \quad (47)$$

where $W_l, b_l \in \theta$ represent the weight matrices and bias vectors of the l th ($l = 1, \dots, L$) hidden layer, respectively; $\text{LReLU}(\cdot)$ is the Leaky rectified linear unit function. The architecture of the designed neural network (policy network) is shown in Fig. 3.

As comparison, pure electric system, i.e., electric heating and cooking devices, is considered in this paper. Other devices of the pure electric system are the same as those of the HMES. For the pure electric system, the action of the agent is $a_t = [c_{WM}^t, c_{IC}^t, P_{RH}^t, P_{AC}^t]$. Therefore, for the designed neural network (policy network) in this case, its output layer has 2 sigmoid neurons and 4 linear neurons. The 2 sigmoid neurons output the probability for WM and IC scheduling, and the 4 linear neurons output the mean and logarithmic standard deviation for RH and AC scheduling (as indicated by the dotted boxes in Fig. 3).

3.3. Proximal policy optimization-based approach

In this subsection, a policy gradient algorithm is used to train the policy network designed in Section 3.2. Policy gradient algorithms have

made great progress in recent years [31]. Among them, trust domain policy optimization (TRPO) and PPO are capable of handling both continuous and discrete action tasks and are well suited for solving the MDP formulated in this study. PPO takes advantage of the reliable performance of TRPO and reduces the computational complexity, which is much simpler to implement and less sensitive to parameter tuning [31]. PPO has become the default DRL algorithm in OpenAI due to its comparable or even better performance than other algorithms. Therefore, we apply PPO to implement the training of the policy network in this paper.

PPO is based on actor-critic structure, which takes stochastic policy network as actor and state-value network as critic. PPO constructs an unconstrained surrogate objective function to remove the incentive for large policy updates. The objective $L^{CLIP}(\theta) \approx J(\pi_\theta)$ is used to calculate the policy gradient. The objective (surrogate objective) is defined as.

$$L^{CLIP}(\theta) = \mathbb{E}_t[\min(\psi_t(\theta) \hat{A}_t, \text{clip}(\psi_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t)] \quad (48)$$

where $\hat{A}_t$ is an estimator of the advantage function at time step t , which represents the advantage of taking an action a_t at a given state s_t ; the clip function $\text{clip}(\psi_t(\theta), 1 - \epsilon, 1 + \epsilon)$ constrains the value of ψ_t to stay within a small interval $[1 - \epsilon, 1 + \epsilon]$; ϵ is a hyper-parameter to control the clipping degree; $\psi_t(\theta)$ is the policy probability ratio which can be interpreted as a measure of the similarity between the updated new policy π_θ and the old policy $\pi_{\theta_{old}}$, defined as.

$$\psi_t(\theta) = \frac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{old}}(a_t | s_t)} \quad (49)$$

In PPO, the generalized advantage estimation (GAE) is employed to make a compromise between bias and variance. The GAE function is applied to calculate the advantage value and written as [32].

$$\hat{A}_t = (1 - \lambda)(\hat{A}_t^{(1)} + \lambda \hat{A}_t^{(2)} + \lambda^2 \hat{A}_t^{(3)} + \dots) \quad (50)$$

where λ is used to adjust the bias-variance trade-off, and $\hat{A}(k)$ is defined as.

$$\hat{A}_t^{(k)} = -V(s_t) + r_t + \gamma r_{t+1} + \dots + \gamma^{k-1} r_{t+k-1} + \gamma^k V(s_{t+k}) \quad (51)$$

where $V(s_t)$ is the state-value function, defined as.

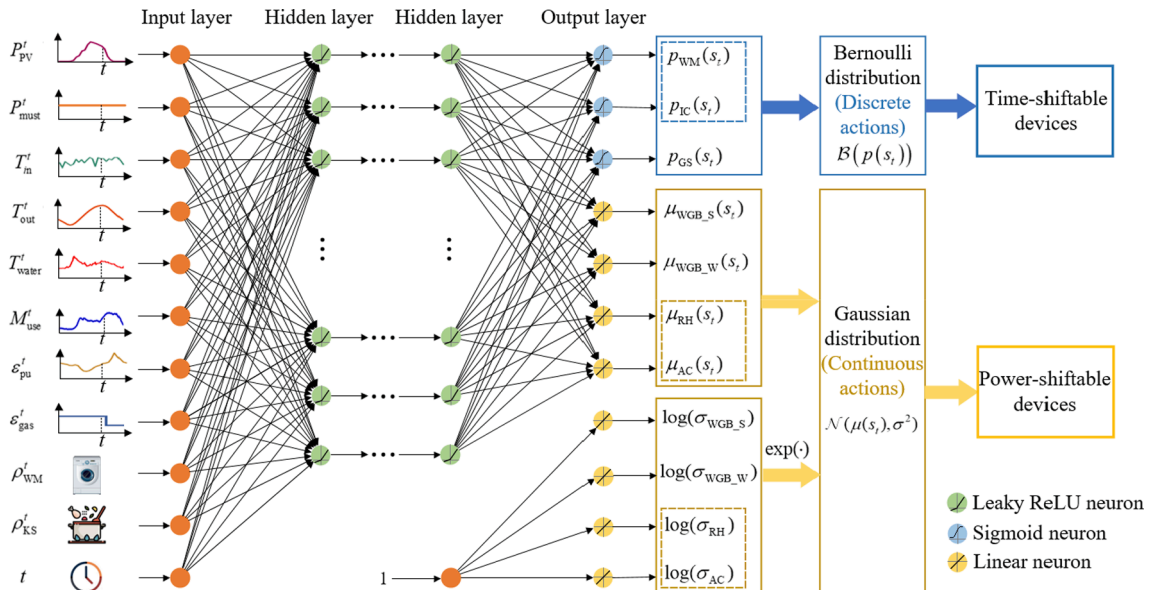


Fig. 3. The architecture of the designed neural network.

$$V(s_t) = \mathbb{E}_{a_t, s_{t+1}, \dots} \left[\sum_{j=0}^{\infty} \gamma^j r_{t+j} \right] \quad (52)$$

Define the time difference residual δ_t as.

$$\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t) \quad (53)$$

Then, the generalized advantage estimator can be written as:

$$\hat{A}_t = \sum_{j=0}^{\infty} (\gamma \lambda)^j \delta_{t+j} \quad (54)$$

The neural network architecture shares same parameters between the policy and value function. Both of the policy network and value network take the environment state as input but are with different outputs. For the policy network for the HMES, the output layer outputs the probability distribution of all actions. Specifically, the output layer has 3 sigmoid neurons and 8 linear neurons. The 3 sigmoid neurons output the probability for WM, IC and GS scheduling, and the 8 linear neurons output the mean and logarithmic standard deviation for WGB_Space, WGB_Water, AC and RH scheduling. For the value network, the output layer has 1 linear neuron which outputs the value of the state. The overall objective is a combination of the error terms of the policy surrogate and the loss function for the value function approximation [32]. The policy surrogate can be derived by substituting Eq. (49) and Eq. (54) into Eq. (48), and the loss function can be defined as.

$$L^{VF}(\theta) = \mathbb{E}_t \left[V(s_t) - \sum_{j=0}^{\infty} \gamma^j r_{t+j} \right]^2 \quad (55)$$

Combining these terms, the overall objective function can be defined as [32].

$$L^{PPO}(\theta) = L^{CLIP}(\theta) - c \cdot L^{VF}(\theta) \quad (56)$$

where c is a loss coefficient.

Finally, the parameters θ are updated by maximizing the overall objective function.

$$\theta \leftarrow \theta + \alpha \nabla_{\theta} L^{PPO}(\theta) \quad (57)$$

where ∇ denotes a gradient calculation, α is learning step-size parameter. The detailed training procedure of the PPO-based HMES scheduling strategy is presented in Algorithm 1.

Algorithm 1. The proposed PPO-based HMES scheduling strategy

- 1: Initialize the deep neural network parameters θ randomly to obtain π_{θ} .
- 2: Initialize the sampling policy $\pi_{\theta_{old}}$ with $\theta_{old} \leftarrow \theta$ and initialize the hyper-parameters.
- 3: **for** iteration $i = 1$ to I **do**
- 4: Receive the initial observation state s_1 of the household multi-energy system.
- 5: **for** $t = 1$ to T **do**
- 6: Sample an energy scheduling action a_t at a given state s_t according to $\pi_{\theta_{old}}$.
- 7: Take action a_t and calculate the reward r_t in Eq. (37).
- 8: Transit to the next state s_{t+1} .
- 9: Form the trajectory $\{s_1, a_1, r_1, s_2, a_2, r_2, \dots, s_b, a_b, r_b\}$.
- 10: **end for**
- 11: Calculate the advantage estimates of $\hat{A}_0, \hat{A}_1, \dots, \hat{A}_T$ using Eq. (54).
- 12: Optimize $L^{PPO}(\theta)$ with respect to θ via Adam optimizer.
- 13: $\theta_{old} \leftarrow \theta$.
- 14: **end for**
- 15: **return:** optimal scheduling policy.

After the completion of the offline training process, the weights of the neural networks will be fixed. Then, the learned policy can be used for HMES decision-making. To be specific, the trained policy network samples action a_t based on the current HMES observation s_t , i.e., PV generation, demand of inflexible loads, indoor temperature, outdoor temperature, hot water temperature, hot water demand, electricity price, gas price, the progresses of laundry task and cooking task. Next,

the action is executed by WM, IC, GS, WGB, RH and AC. The agent receives reward r_t and the environment transits to the next state. Then, the agent observes the state s_{t+1} as a new state, and makes the energy scheduling decision of this time step. The procedure repeats until the last time step of the day. It can be seen from this procedure that the decision-making of the developed method requires no prediction or distribution knowledge of system uncertainties. Therefore, the unreliability of the schedules caused by inaccurate prediction or modeling can be avoided.

4. Case studies

4.1. Experimental setup

The HMES shown in Fig. 1 is used for case studies in this paper. Parameters regarding space and water heating (including WGB, AC, RH and other thermal related parameters) are listed in Table 2. Operational parameters of WM and KS are listed in Table 3. The weighting factors w_1 , w_2 and w_3 are set as 2, 1 and 1, respectively. It is noted that the weighting factors can be set to other values according to different user preferences. The PV generation and hot water demand data are generated based on the open source CREST (Center for Renewable Energy Systems Technology) model [33], which has been widely used [34,35]. The data of outdoor temperature is adopted from [36]. The data of electricity purchasing price is adopted from [37], and the selling price is 0.5 times that of the purchasing price. The natural gas price is 33.81 cents/m³ during 0:00–20:00 and 14.8 cents/m³ at all other times. The electric power demand of non-shiftable loads (REFG and AS) is adopted from [38] and set as 0.6 kW throughout the day. In the simulation, data from Dec. 1, 2017 to Feb. 28, 2018 are used for training and data from Dec. 1, 2018 to Feb. 28, 2019 are used for performance testing. The simulation is conducted for a 24-h horizon and the simulation step is 15 min.

The proposed PPO-based joint scheduling approach is implemented using Python. The hyper-parameters of PPO are chosen according to the common practice [39] and tuned by trial-and-error. The policy network (actor) and value network (critic) both have 3 hidden layers and each layer has 128 neurons. The discount factor is set as 0.99, the bias-

Table 2
Parameters regarding space and water heating.

Device	Parameter	Value
WGB	Q_{rad}^{unit}	1.12 kW
	n	6
	M_{med}	30 kg
	C_{med}	4.2 kJ/(kg·°C)
	ΔT_s	64.5°C
	$T_{WGB-set}$	55°C
	T_{cold}	10°C
	M_{tank}	100 kg
	η_{WGB}	0.89
	$Q_{WGB_Space}^{max}$	6 kW
	$Q_{WGB_Water}^{max}$	6 kW
	T_{min}^{med}	5°C
	T_{max}^{med}	85°C
	P_{AC}^{max}	2 kW
AC	η_{AC}	3.4
RH	P_{RH}^{max}	3.6 kW
	η_{RH}	0.98
House	R	18 °C/kW
	C	0.525 kWh/°C
	T_{in}^{pre}	23°C
	T_{water}^{pre}	50°C
Residents	T_{in}^{min}	21°C
	T_{in}^{max}	25°C
	T_{water}^{min}	45°C
	T_{water}^{max}	60°C
	C_{water}	4.2 kJ/(kg·°C)
	H_{gas}	9.8 kWh/m ³
Others	σ_1/σ_2	0.15/0.25 cents

Table 3
Operational parameters of WM and KS.

Device	Rated power (kW)	Starting time	Deadline	Required time (h)
WM	1	10:00	22:00	1.5
KS	2	6:00	9:00	1
		11:00	13:00	1
		17:00	20:00	1

variance trade-off parameter is 0.95, and the clip parameter is 0.1. The loss coefficient is 0.5, and the learning step-size parameter is 0.001. Adam is used to stabilize the gradient direction.

4.2. Performance evaluation

Different cases are designed to assess the performance of the proposed PPO-based load scheduling approach. Specifically, we analyze the obtained HMES scheduling results of two typical winter days (a weekday and a weekend) selected in the test set. The two days represent different hot water consumption patterns in weekday and weekends and different PV outputs and outdoor temperatures due to different weather conditions. As comparison, pure electric system is considered. Therefore, four cases are designed in this study.

Case 1: HMES (weekday).

The HMES shown in Fig. 1 is used for performance evaluation in this case. The profiles of PV generation, hot water demand, electricity price and outdoor temperature are shown in Fig. 4. The scheduling results of Case 1 are shown in Fig. 5.

The scheduling actions obtained by the PPO-based approach, i.e., scheduling results of Case 1 are shown in Fig. 5. It can be observed from Fig. 5(a) that the hot water temperature is always within the range of [45, 60] °C. The water heating is mainly carried out by WGB. At the time of 10:00, 11:00, 11:45, etc, the electricity price is low and RH performs water heating to reduce the cost of hot water production. Especially at the time of 11:45, when the electricity price is at the lowest value of the day, RH works at a higher power, making the water temperature rise rapidly from the minimum allowed value of 45 °C to 50.86 °C. As shown in Fig. 5(b), the indoor temperature is well controlled within [21, 25] °C. Space heating is mainly provided by AC. Since the electricity price is at the highest value during 17:00–18:00, AC is not working during this period and the indoor temperature rapidly decreases from 25 °C to 22.19 °C. The natural gas price is low during 20:00–21:15, WGB consumes gas for space heating. The PV output first serves the power demand of the non-shiftable loads, and the surplus onsite generation provides power for other electrical loads. For the time-shiftable devices WM and IC, it can be seen from Fig. 5(c) that they both work during the low tariff periods to reduce the energy cost. For the cooking task of dinner, it is done by GS due to the higher electricity price. This indicates that the proposed approach is able to jointly schedule each household device and perform energy switching to minimize the energy cost and the thermal comfort loss. The amount of gas consumption of this case is 6.54 m³. The total energy cost of this case is 168.31 cents, the penalty caused by hot water temperature discomfort is 4.73 cents, and the

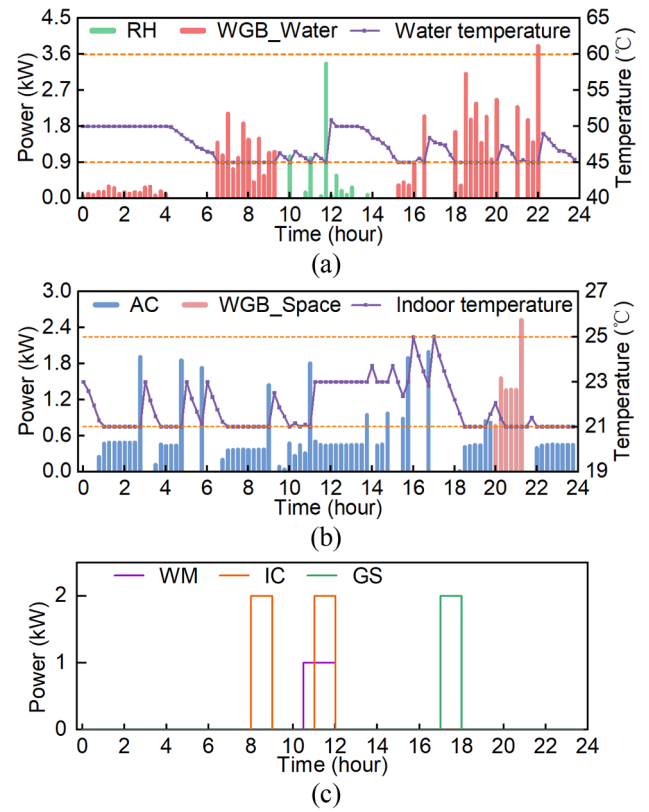


Fig. 5. Scheduling results of Case 1: (a) water heating; (b) space heating; (c) time-shiftable devices.

penalty caused by indoor temperature discomfort is 4.71 cents.

Case 2: Pure electric system (weekday).

The system with pure electric heating and cooking is considered in this case. The profiles of PV generation, hot water demand, electricity price and outdoor temperature are the same as those in Case 1. The scheduling results of Case 2 are shown in Fig. 6.

As shown in Fig. 6(a) and 6(b), the hot water temperature and indoor temperature are maintained within the defined ranges. Since all the devices in this case are electrical loads, they no longer have the capability of “energy shift” compared to Case 1, and the water heating is done by RH which is more costly. When the hot water demand is large and the electricity price is high (such as 18:00–20:00), RH works continuously to maintain the hot water temperature at the minimum allowable value (45 °C), thus reducing the energy cost while maintaining users’ thermal comfort. AC operates similarly to the operation in Case 1, such as switching off when the electricity price is extremely high. At the time of 11:00, when the electricity price is at the lowest value, AC works at a higher power, making the water temperature rise rapidly from the minimum allowed value of 21 °C to 22.89 °C. It can be seen from Fig. 6 (c) that WM and IC both operate during the low electricity price periods

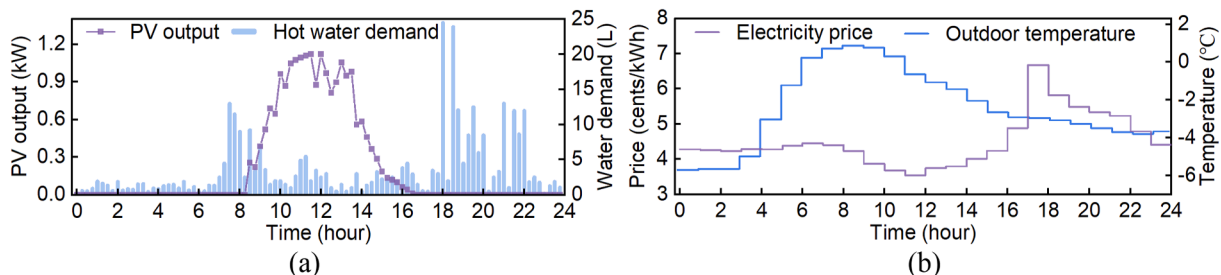


Fig. 4. The profiles of winter weekday: (a) PV output and hot water demand; (b) electricity price and outdoor temperature.

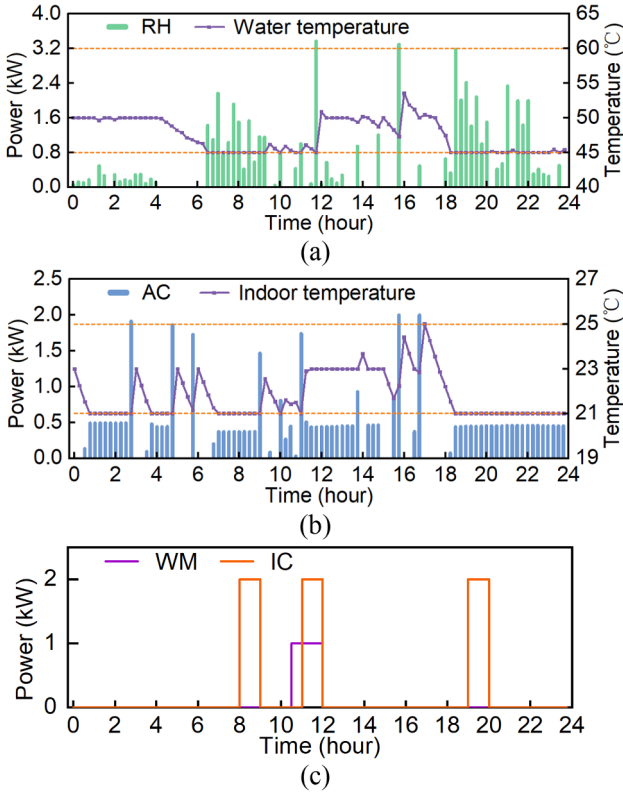


Fig. 6. Scheduling results of Case 2: (a) water heating; (b) space heating; (c) time-shiftable devices.

to minimize the energy cost. The total energy cost of this case is 189.58 cents, which is 12.64% higher than that of Case 1. The penalty costs for hot water temperature discomfort and indoor temperature discomfort are 4.41 cents and 4.77 cents, respectively.

Case 3: HMES (weekend).

In this case, the system composition is the same as that in Case 1. The profiles of this case are shown in Fig. 7. The scheduling results of Case 3 are shown in Fig. 8.

Compared to Case 1 (weekday), there is a larger hot water demand during 10:00–12:30 in this case. Therefore, there is always water heating during this period to meet users' hot water demand. At the same time, the water temperature in this stage is always at the minimum allowable value (45 °C) in order to reduce the cost of water heating. It can be seen that the PV output is large during 11:45–12:30, and the PV output first meets the demand of non-shiftable loads, and the surplus generation provides power for RH water heating; due to the high electricity price in this phase, WGB is used as a supplement to heat the water. As shown in Fig. 8(b), the space heating is mainly provided by AC. When the natural gas price is low, WGB consumes gas for space heating, such as the period during 20:00–22:00. In Fig. 8(c), WM is scheduled to operate during the periods when the electricity price is relatively low in

its predefined working time. The breakfast is prepared by IC, while the lunch and dinner are prepared by GS. The amount of gas consumption of the case is 6.43 m³. The total energy cost of the case is 167.24 cents, the penalty caused by hot water temperature discomfort is 4.50 cents, and the penalty caused by indoor temperature discomfort is 3.61 cents.

Case 4: Pure electric system (weekend).

In this case, the system composition is the same as that in Case 2. The profiles of PV generation, hot water demand, electricity price and outdoor temperature are the same as those in Case 3. The scheduling results of Case 4 are shown in Fig. 9.

In Case 4, the PPO-based approach also realizes the joint scheduling of power-shiftable devices (AC and RH) with continuous actions and time-shiftable devices (WM and IC) with discrete actions. The hot water temperature and indoor temperature are maintained within the user-defined ranges. Due to the high electricity price during 11:00–13:00, both RH and AC work at high power during 10:00–10:45 to reduce the operation cost, with the hot water temperature and indoor temperature rising rapidly during this period. The total energy cost of this case is 186.62 cents, an increase of 11.59% compared to Case 3. The penalty costs for hot water temperature discomfort and indoor temperature discomfort are 4.12 cents and 3.35 cents, respectively. Simulation results show that the PPO-based approach can continuously optimize the operation of each device to minimize the electricity cost and thermal comfort loss.

4.3. Benefit of joint discrete and continuous decision-making

To illustrate the advantage of the proposed approach, i.e., the ability to jointly schedule different types of loads in the HMES hybrid action space, we compare the scheduling results of the proposed approach with the DQN-based approach, which is a widely known DRL method. For the DQN network, its input is the state and its outputs are Q-value estimates for all actions. It needs to discretize the continuous actions of power-shiftable devices. Considering that the agent's action contains decision variables of multiple devices, the action dimension will increase greatly with the granularity of discretization, which increases the difficulty of solving the problem. In this regard, $Q_{WGB_Space}^t$, $Q_{WGB_Water}^t$, P_{RH}^t , and P_{AC}^t are discretized into 2 integer values, respectively. That is, the optional actions of WGB space heating and water heating are [0, 6] kW, the actions of RH are [0, 3.6] kW and the optional actions of AC are [0, 2] kW. We select Case 1 of Section 4.2 for performance testing. The DQN network has 3 hidden layers. The discount factor is 0.99 and the learning rate is 0.001.

The scheduling results obtained by the DQN-based approach are shown in Fig. 10. Since the actions of WGB, AC and RH need to take the set discrete values, the feasible action options are greatly reduced and the flexibility of power-shiftable devices cannot be effectively utilized. The variation range of the hot water temperature is much larger than that in Fig. 5 (a), and the fluctuation frequency of the indoor temperature is more obvious than that in Fig. 5 (b). This is because the discretization operation of the action space limits the flexibility of the devices. For discrete actions, the working condition of cooking task is consistent with that in Fig. 5 (c), and the WM works during the period of

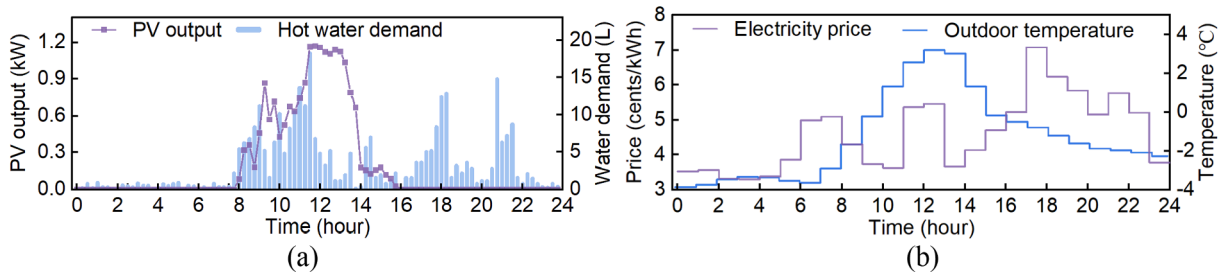


Fig. 7. The profiles of winter weekend: (a) PV output and hot water demand; (b) electricity price and outdoor temperature.

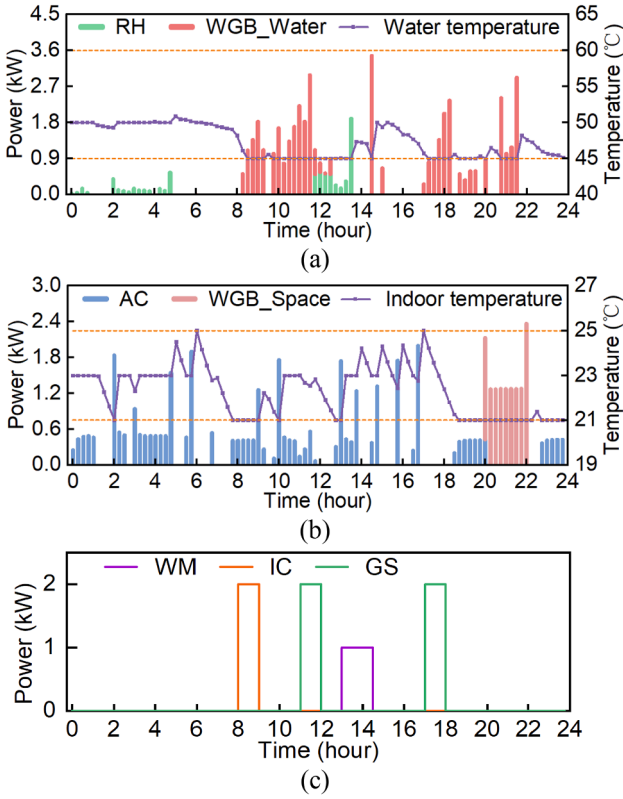


Fig. 8. Scheduling results of Case 3: (a) water heating; (b) space heating; (c) time-shiftable devices.

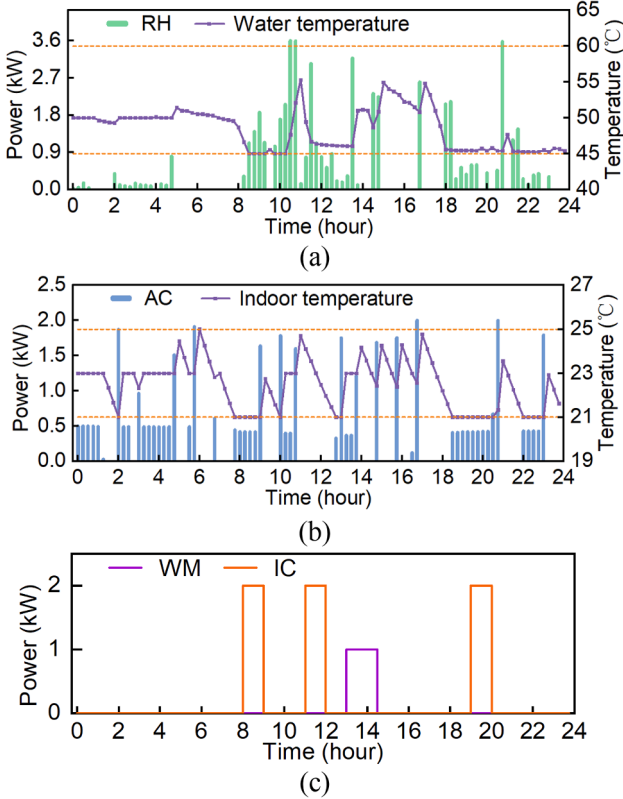


Fig. 9. Scheduling results of Case 4: (a) water heating; (b) space heating; (c) time-shiftable devices.

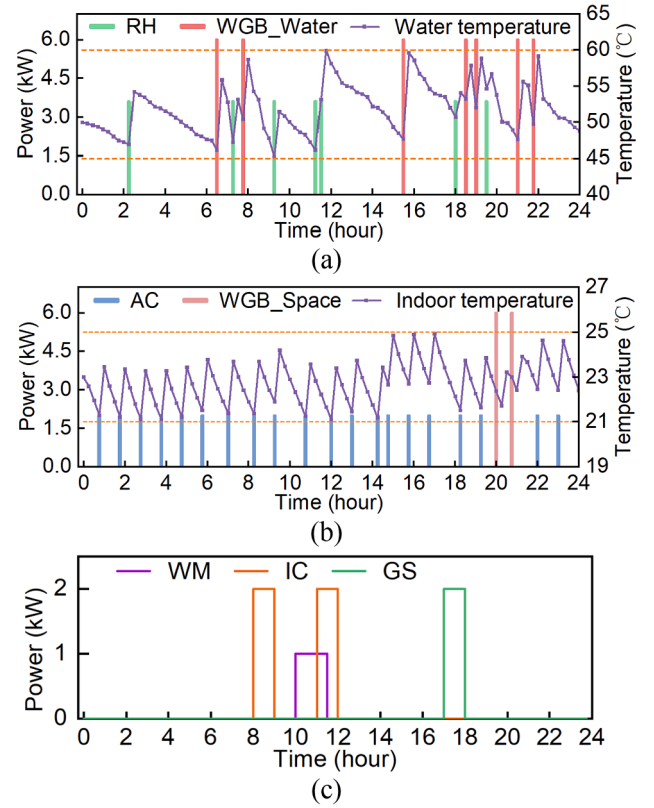


Fig. 10. Scheduling results of Case 1 based on DQN.

low electricity price and high PV output. The total energy cost based on DQN is 190.53 cents, the penalty caused by hot water temperature discomfort is 4.87 cents, and the penalty caused by indoor temperature discomfort is 2.95 cents. Despite the indoor temperature discomfort is less than that of the PPO-based approach, the total energy cost increases by 13.2% and the hot water temperature discomfort also increases. It can be concluded that discretization of continuous actions will lead to sub-optimal HMES scheduling strategy. Therefore, the PPO-based scheduling approach proposed in this paper is more suitable for solving the load scheduling problem of HMES. It can design cost-effective scheduling decisions by exploring the whole continuous action space of power-shiftable devices and the discrete action space of time-shiftable devices respectively.

4.4. Comparison with other approaches

In this subsection, a comparison is conducted among the proposed approach and an MPC-based scheduling approach and a DQN-based approach to further verify the superiority of the proposed PPO-based HMES scheduling approach.

- (1) MPC: It needs to predict the PV generation, outdoor temperature, hot water demand and electricity price first. The load scheduling problem is solved based on the prediction. It uses a fully connected neural network with one hidden layer to predict these information, and the selected rolling horizon is set as 3 h.
- (2) DQN: Its input is the state s_t and the outputs are Q-value estimates. The discretization of the continuous actions $Q_{WGB_Space}^t, Q_{WGB_Water}^t, P_{RH}^t, P_{AC}^t$ is consistent with the discretization operation described in Section 4.3.

The 90 days in the test set are used to compare the daily energy cost and thermal discomfort penalty costs obtained using the approaches described above. The average values of daily energy cost, penalty for hot

Table 4
Statistics of the average daily costs.

Cost (cents)	Energy cost	Penalty for hot water temperature discomfort	Penalty for indoor temperature discomfort
The proposed (Electric)	181.06	4.33	3.71
The proposed (HMES)	157.33	4.27	3.64
DQN (HMES)	179.12	4.56	3.02
MPC (HMES)	164.90	4.41	3.73

water temperature discomfort and penalty for indoor temperature discomfort are given in Table 4. Besides, in order to illustrate the economic benefits of the hybrid energy solution over the pure electric solution, the statistical results of the proposed PPO-based approach on pure electric system are also given in Table 4.

As observed from Table 4, the gas-electric HMES can save 13.11% on average daily energy cost compared to the pure electric solution using the PPO-based approach. The proposed approach achieves better performance on the test set and can provide cost-effective and comfort-aware HMES scheduling solution. Despite the DQN-based approach achieves lower indoor temperature discomfort, it significantly increases the energy cost due to the inflexibility of DQN to control WGB, AC, and RH. And the discretization operation of continuous action results in a sub-optimal scheduling strategy. The average daily energy cost of the proposed approach is reduced by 12.17% compared to the DQN approach and 4.59% compared to the MPC approach. Due to the uncertainties of users' demand, PV output, outdoor temperature and electricity price, the HMES scheduling approach based on prediction is limited by the prediction accuracy, which makes it difficult to accurately adapt to the dynamic changes of the actual environment. The PPO-based joint load scheduling approach proposed in this paper has better implementability by avoiding the dependence on knowledge of system uncertainty distribution and prediction information.

4.5. Sensitivity analysis

In this subsection, sensitivity analysis based on four uncertainties is conducted to assess the impact of the change of relevant factors on the operation performance (energy costs) of the HMES based on the proposed scheduling approach. The sensitivities of energy cost to electricity price, PV generation, hot water demand and outdoor air temperature are assessed. When assessing one factor, the factor varied from 70% to 130% (70%, 80%, ..., 130%) of the value defined in Case 1 (Fig. 4), and all the other factors remained unchanged as defined in Case 1. The energy costs are shown in Fig. 11.

As shown in Fig. 11, the energy costs of the HMES increase with the increase of electricity price and hot water demand. With the growth of outdoor temperature and PV generation, the energy costs of the system show a slow downward trend. The energy cost increases significantly with the increase of electricity price. When outdoor temperature

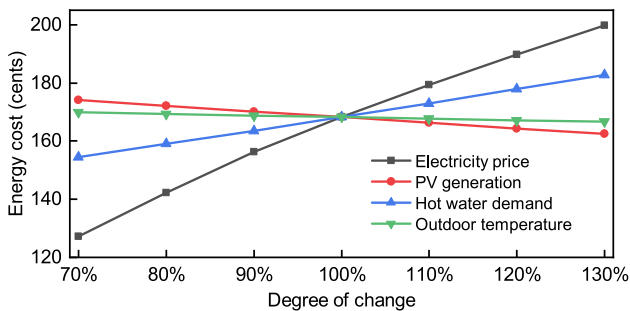


Fig. 11. Sensitivity analysis of electricity price, PV generation, hot water demand and outdoor temperature.

changes, the energy cost changes slowly. When the electricity price increases by 30%, the energy cost increases by 18.7%; when the electricity price decreases by 30%, the energy cost decreases by 24.43%. When the outdoor temperature increases by 30%, the energy cost decreases by 0.99%; when the temperature decreases by 30%, the energy cost increases by 0.9%. This is mainly due to the small base of the outdoor temperature and the small impact on energy cost when the change is small.

In addition, the operation costs of the four cases considering and ignoring the proposed approach are compared. When ignoring the proposed approach, the load scheduling is achieved through a rule-based approach. Specifically, the rule is defined as follows: when the indoor temperature is below 22 °C, the heating system will start work at maximum power for space heating, otherwise it will be closed; when the hot water temperature is below 48 °C, the heating system will start work at maximum power for water heating, otherwise it will be closed; the time-shiftable devices start working at the predefined starting time. The operation costs of the four cases based on the defined rule are given in Table 5.

Compared to the energy cost obtained by using the proposed approach given in Section 4.2, the energy cost obtained by using the rule-based approach increases by 13.8%, 12.66%, 12.65% and 13.26% in Case 1, Case 2, Case 3 and Case 4 respectively. The comparison results verify the effectiveness of the proposed approach in terms of cost saving.

5. Conclusion

In this paper, we propose a deep reinforcement learning-based joint load scheduling approach for household multi-energy system (HMES) operation optimization, which takes into account the uncertainties of electricity price, renewable generation, residents' heat demand and outdoor temperature. Specifically, the studied energy cost and thermal discomfort minimization problem is converted to a Markov decision process, and solved using a proximal policy optimization approach. Different from traditional approaches that rely on prediction or distribution knowledge of system uncertainties, the proposed approach can optimally schedule the usage of different household devices only based on the instantaneous observation. Therefore, the proposed approach has better implementability and superiority than traditional approaches, especially in the case of inaccurate or even missing prediction information. It can simultaneously handle the continuous actions of wall-hung gas boiler, air conditioner and resistance water heater, as well as the discrete actions of kitchen stove and washing machine. In addition, it can optimize the operation of electric-gas devices to reduce the energy cost, which is well suited for the HMES environment where there are multiple different types of household loads and a large combination of working conditions. The proposed approach also has some limitations: it needs large amounts of historical data and takes a long time for offline training.

The results of case studies show that the proposed approach can dynamically optimize the operation of each device to reduce the energy cost and minimize the thermal discomfort, while accommodating the uncertainties of hot water demand, electricity price, PV generation and outdoor temperature. Compared to the pure electric solution, the gas-electric HMES can provide more economical and flexible energy supply solution for smart homes, saving 13.11% on average daily energy

Table 5
Operation costs of the four cases based on the rule-based approach.

Cost (cents)	Energy cost	Penalty for hot water temperature discomfort	Penalty for indoor temperature discomfort
Case 1	191.53	5.05	2.84
Case 2	213.59	3.74	2.72
Case 3	188.39	4.54	3.01
Case 4	211.37	3.42	2.68

cost. The advantage of the proposed approach in being able to jointly make discrete and continuous decisions is demonstrated by comparing the results with the DQN-based approach. In addition, the proposed approach is compared to an MPC-based scheduling approach to further verify the superiority of the proposed approach in terms of cost saving and comfort guarantee. The average energy cost of the proposed approach is reduced by 12.17% compared to the DQN-based approach and 4.59% compared to the MPC-based approach.

The proposed approach can adapt to larger formulations of HMES including power-shiftable devices (such as heat pump) and time-shiftable devices (such as dishwasher). Since the proposed approach can jointly schedule a variety of household devices with different operation characteristics, our solution provides a good basis for expanding its application in larger scales.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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